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ELECTRONIC DEVICE FOR CONTROLLING TRANSMISSION POWER, AND ELECTRONIC DEVICE OPERATING METHOD

Abstract

An electronic device is provided. The electronic device includes a temperature sensor, an application processor, a radio frequency (RF) module, memory, comprising one or more storage media, storing instructions, and one or more communication processors communicatively coupled to the temperature sensor, the application processor, the RF module, and the memory, wherein the instructions, when executed by the one or more communication processors individually or collectively, cause the electronic device to acquire, from the application processor, information related to the electronic device temperature measured by the temperature sensor, determine a parameter related to the transmission power of the RF module on the basis of the electronic device temperature, and control the RF module on the basis of the determined parameter.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365 (c), of an International application No. PCT/KR2023/014810, filed on Sep. 26, 2023, which is based on and claims the benefit of a Korean patent application number 10-2022-0161967, filed on Nov. 28, 2022, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2022-0187470, filed on Dec. 28, 2022, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The disclosure relates to an electronic device for controlling transmission power and an electronic device operating method.

2. Description of Related Art

[0003] In electronic devices to which 5th generation (5G) new radio (NR) is applied, there is a high possibility that the electronic device will generate heat due to situations such as the use of a high frequency band and the use of a wide bandwidth compared to long term evolution (LTE). The increase in the number of cameras and the installation of high-definition displays also cause heat generation in the electronic device, which can lead to the risk of low-temperature burns. Therefore, the need for heat generation control on the basis of the temperature of the electronic device may be required.

[0004] Among the technologies of limiting heat generation, there may be a method of reducing transmission power in communication. For example, when the electronic device generates heat greater than or equal to a designated value, NR call drop and/or LTE fallback may be performed to prevent the temperature of the electronic device from rising. In this case, the radio frequency (RF) module may not operate normally, making it difficult to perform communication seamlessly.

[0005] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

[0006] Time averaged specific absorption rate (SAR) (TAS) refers to a time-dependent average specific absorption rate (SAR), and may be a technology for controlling a communication module so that the electronic device controls the averaged SAR within a target SAR range for a period defined in a standard.

[0007] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide an electronic device for controlling transmission power of an RF module in response to the temperature exceeding a designated value. More particularly, the electronic device adjusts a threshold value (Plimit) of the transmission power for satisfying the target SAR for a designated period of time in relation to TAS control in response to the temperature exceeding the

designated value, a value (Preserve) for outputting the designated power to lower the transmission power when the transmission power exceeds the designated value, and/or maximum power (Pmax) that a transmitter of the RF module may output, so communication may be performed seamlessly even if the temperature of the electronic device rises.

[0008] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

[0009] In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes a temperature sensor, an application processor, a radio frequency (RF) module, memory, comprising one or more storage media, storing instructions, and one or more communication processors communicatively coupled to the temperature sensor, the application processor, the RF module, and the memory, wherein the instructions, when executed by the one or more communication processors individually or collectively, cause the electronic device to: acquire, from the application processor, information related to the temperature of the electronic device measured by the temperature sensor, determine a parameter related to the transmission power of the RF module on the basis of the temperature of the electronic device, and control the RF module on the basis of the determined parameter.

[0010] In accordance with an aspect of the disclosure, a method of operating an electronic device is provided. The method includes acquiring information related to a temperature of the electronic device measured by a temperature sensor from an application processor, determining a parameter related to transmission power of an RF module on the basis of the temperature of the electronic device, and controlling the RF module on the basis of the determined parameter.

[0011] For example, the electronic device performs communication seamlessly by controlling the transmission power even if the temperature rises due to the heat generation of the electronic device.

[0012] In accordance with an aspect of the disclosure, one or more non-transitory computer-readable storage media storing one or more computer programs including computer-executable instructions that, when executed by one or more communication processors of an electronic device individually or collectively, cause the electronic device to perform operations is provided. The operations include acquiring information related to a temperature of an electronic device measured by a temperature sensor from an application processor, determining a parameter related to transmission power of an radio frequency (RF) module based on the temperature of the electronic device, and controlling the RF module based on the determined parameter.

[0013] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0015] FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure;

[0016] FIG. 2 is a block diagram of an electronic device according to an embodiment of the disclosure;

[0017] FIG. 3 is a time-dependent transmission power graph related to time averaged SAR (TAS) according to an embodiment of the disclosure;

[0018] FIG. 4 is a flowchart illustrating a method for a communication processor to control a radio frequency (RF) module on a basis of a temperature of an electronic device according to an

embodiment of the disclosure;

[0019] FIG. 5A is a flowchart illustrating a method for an electronic device to control an RF module on a basis of a temperature of an electronic device according to an embodiment of the disclosure;

[0020] FIG. 5B is a diagram illustrating a transmission power graph when an electronic device controls an RF module on a basis of a temperature of an electronic device according to an embodiment of the disclosure;

[0021] FIG. 6A is a flowchart illustrating a method for an electronic device to control an RF module on a basis of a temperature of an electronic device according to an embodiment of the disclosure;

[0022] FIG. 6B is a diagram illustrating a transmission power graph when an electronic device controls an RF module on a basis of a temperature of an electronic device according to an embodiment of the disclosure;

[0023] FIG. 7A is a flowchart illustrating a method for an electronic device to control an RF module on a basis of a temperature of an electronic device according to an embodiment of the disclosure; and

[0024] FIG. 7B is a diagram illustrating a transmission power graph when an electronic device controls an RF module on a basis of a temperature of an electronic device according to an embodiment of the disclosure.

[0025] Throughout the drawings, it should be noted that like reference numbers are used to depict the same or similar elements, features, and structures.

DETAILED DESCRIPTION

[0026] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

[0027] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

[0028] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0029] It should be appreciated that the blocks in each flowchart and combinations of the flowcharts may be performed by one or more computer programs which include instructions. The entirety of the one or more computer programs may be stored in a single memory device or the one or more computer programs may be divided with different portions stored in different multiple memory devices.

[0030] Any of the functions or operations described herein can be processed by one processor or a combination of processors. The one processor or the combination of processors is circuitry performing processing and includes circuitry like an application processor (AP, e.g. a central processing unit (CPU)), a communication processor (CP, e.g., a modem), a graphics processing unit (GPU), a neural processing unit (NPU) (e.g., an artificial intelligence (AI) chip), a Wi-Fi chip, a Bluetooth® chip, a global positioning system (GPS) chip, a near field communication (NFC) chip, connectivity chips, a sensor controller, a touch controller, a finger-print sensor controller, a display

driver integrated circuit (IC), an audio CODEC chip, a universal serial bus (USB) controller, a camera controller, an image processing IC, a microprocessor unit (MPU), a system on chip (SoC), an IC, or the like.

[0031] FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to an embodiment of the disclosure.

[0032] Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). In an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**.

According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In various embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0033] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation.

According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to another embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a designated function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0034] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). In an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to another embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure designated for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural

network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0035] The memory **130** may be configured to store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0036] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0037] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0038] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. In an embodiment, the speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0039] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to another embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0040] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0041] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. In an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0042] The interface **177** may support one or more designated protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to another embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0043] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). In an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0044] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile

sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0045] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0046] The power management module **188** may manage power supplied to the electronic device **101**. The power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0047] The battery **189** may supply power to at least one component of the electronic device **101**. In an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0048] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to another embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0049] The wireless communication module **192** may support a 5G network, after a 4.sup.th generation (4G) network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the millimeter wave (mmWave) band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may, for example, support various requirements designated in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

[0050] The antenna module **197** may transmit or receive a signal or power to or from the outside

(e.g., the external electronic device) of the electronic device **101**. According to another embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). The antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

[0051] According to some embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mm Wave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0052] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0053] In an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to another embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) on the basis of 5G communication technology or IoT-related technology.

[0054] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment, the electronic devices are not limited to those described above.

[0055] It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0056] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0057] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0058] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0059] According to some embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to other embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various

embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added. [0060] FIG. 2 is a block diagram of an electronic device **200** according to an embodiment of the disclosure.

[0061] Referring to FIG. 2, an electronic device **200** (e.g., an electronic device (**101**) of FIG. 1) may include an application processor **210** (e.g., an processor **120** of FIG. 1), a temperature sensor **260**, a communication processor **220**, and/or an RF module **230**. The components included in FIG. 2 are for some of the components included in the electronic device **200**, and the electronic device **200** may include various other components as illustrated in FIG. 1.

[0062] The temperature sensor **260** according to various embodiments may measure the temperature of the electronic device **200**. In an example, the temperature sensor **260** is a component for converting temperature into an electrical characteristic value, and may include at least one of a thermistor, a resistance thermometer, a thermoelectric, a silicon transducer, or a critical temperature resistor (CTR). The temperature sensor **260** may include at least one of various other temperature detection elements.

[0063] The application processor **210** according to various embodiments may acquire information related to the temperature of the electronic device from the temperature sensor **260** and transmit the acquired information to the communication processor **220**.

[0064] In an embodiment, the application processor **210** may request information related to the temperature of the electronic device **200** from the temperature sensor **260** and acquire, from the temperature sensor **260**, the information in which the temperature of the electronic device **200** is converted into the electrical characteristic value. For example, the application processor **210** may transmit the information related to the temperature of the electronic device **200** in response to the request of the communication processor **220**.

[0065] The communication processor **220** according to various embodiments may control transmission power of the RF module **230**. For example, the transmission power of the RF module may be controlled. In another example, the communication processor **220** may control the transmission power of the RF module **230** on the basis of the temperature information of the electronic device **200** acquired from the application processor **210**.

[0066] The RF module **230** according to various embodiments may communicate with an external electronic device and/or a server through a wireless communication network. For example, the RF module **230** may transmit and receive information to and from an external electronic device and/or a server through an antenna.

[0067] FIG. 3 is a time-dependent transmission power graph related to time averaged SAR (TAS) according to an embodiment of the disclosure.

[0068] The TAS refers to a time-dependent average specific absorption rate (SAR), and may be a technology for controlling a communication module so that the electronic device **200** controls the averaged SAR within a target SAR range for a period defined in a standard.

[0069] In the TAS control, a parameter related to the transmission power may be applied. For example, the parameter related to the transmission power may include a first parameter, a second parameter, and/or a third parameter.

[0070] For example, the first parameter of the transmission power is a threshold value P_{limit} of the transmission power for satisfying a target SAR for a designated period of time, which may be maximum time-average output power.

[0071] The second parameter of the transmission power may be a value (preserve) for outputting designated power to lower the transmission power when the transmission power exceeds a designated value.

[0072] For example, the third parameter of the transmission power may be maximum power P_{max} that a transmitter of the RF module **230** may output.

[0073] The target SAR may be a reference SAR for satisfying the TAS.

[0074] For example, the averaged SAR may be an averaged SAR for the designated period of time.

[0075] The electronic device **200** may control the RF module **230** on the basis of the first parameter, the second parameter, and/or the third parameter so that the averaged SAR of the transmission power for the designated period of time satisfies the target SAR.

[0076] Time point “A” may be a time point at which the temperature of the electronic device **200** is higher than or equal to the designated value.

[0077] The electronic device **200** may perform an NR Call drop and/or LTE Fallback on the RF module **230** at the time point “A” in response to the temperature sensor **260** detecting that the temperature of the electronic device **200** is higher than or equal to the designated value.

[0078] When the temperature of the electronic device **200** rises higher than or equal to the designated value, the RF module **230** may not operate normally, making it difficult to perform communication seamlessly.

[0079] FIG. **4** is a flowchart illustrating a method for a communication processor **220** to control an RF module **230** on a basis of a temperature of an electronic device **200** according to an embodiment of the disclosure.

[0080] According to various embodiments, in operation **410**, the communication processor **220** may acquire the information related to the temperature of the electronic device **200**.

[0081] According to an embodiment, the temperature sensor **260** may measure the temperature of the electronic device **200**. For example, the temperature sensor **260** is a component for converting temperature into an electrical characteristic value, and may include at least one of a thermistor, a resistance thermometer, a thermoelectric, a silicon transducer, or a critical temperature resistor (CTR). The temperature sensor **260** may include at least one of various other temperature detection elements.

[0082] According to another embodiment, the application processor **210** may acquire the information related to the temperature of the electronic device **200** from the temperature sensor **260**. For example, the application processor **210** may request the information related to the temperature of the electronic device **200** from the temperature sensor **260** and acquire, from the temperature sensor **260**, the information in which the temperature of the electronic device **200** is converted into the electrical characteristic value.

[0083] The communication processor **220** may acquire the information related to the temperature of the electronic device **200** from the application processor **210**. For example, the application processor **210** may transmit the information related to the temperature of the electronic device **200** in response to the request of the communication processor **220**.

[0084] According to various embodiments, in operation **420**, the communication processor **220** may determine the parameter related to the transmission power on the basis of the temperature of the electronic device **200**.

[0085] The parameter related to the transmission power may include the first parameter, the second parameter, and/or the third parameter.

[0086] For example, the first parameter of the transmission power is the threshold value P_{limit} of the transmission power for satisfying the target SAR for the designated period of time, which may be the maximum time-average transmission power.

[0087] The second parameter of the transmission power may be the value for outputting the designated power to lower the transmission power when the transmission power exceeds the designated value.

[0088] In another example, the third parameter of the transmission power may be the maximum power P_{max} that the transmitter of the RF module **230** may output.

[0089] According to an embodiment, the communication processor **220** may determine the second parameter of the transmission power as a first level, a second level, a third level, and/or a fourth level on the basis of the temperature of the electronic device **200**. The first level, the second level,

the third level, and/or the fourth level of the second parameter may be the value related to the transmission power of the designated value for lowering the transmission power.

[0090] In another embodiment, the communication processor **220** may determine the first parameter of the transmission power as the first level, the second level, and/or the third level on the basis of the temperature of the electronic device **200**. The first level, the second level, and/or the third level of the first parameter may be the value related to the average transmission power of the designated value for satisfying the target SAR.

[0091] In an embodiment, the communication processor **220** may determine the third parameter of the transmission power as the designated value on the basis of the temperature of the electronic device **200**. The designated value of the third parameter may be the value related to the transmission power for setting a threshold of the maximum power P_{max} that the transmitter of the RF module **230** may output.

[0092] In operation **430**, the communication processor **220** may control the RF module **230** on the basis of the parameter related to the determined transmission power. For example, the communication processor **220** may control the transmission power by controlling the RF module **230** to satisfy the determined first parameter, second parameter, and/or third parameter.

[0093] FIG. 5A is a flowchart illustrating a method for an electronic device **200** to control an RF module **230** on a basis of a temperature of an electronic device **200** according to an embodiment of the disclosure.

[0094] In operation **510**, the communication processor **220** may acquire the information related to the temperature of the electronic device **200**.

[0095] According to an embodiment, the temperature sensor **260** may measure the temperature of the electronic device **200**. For example, the temperature sensor **260** is a component for converting the temperature into the electrical characteristic value, and may include at least one of the thermistor, the resistance thermometer, the thermoelectric, the silicon transducer, or the critical temperature resistor (CTR). The temperature sensor **260** may include at least one of various other temperature detection elements.

[0096] According to another embodiment, the application processor **210** may acquire the information related to the temperature of the electronic device **200** from the temperature sensor **260**. For example, the application processor **210** may request the information related to the temperature of the electronic device **200** from the temperature sensor **260** and acquire, from the temperature sensor **260**, the information in which the temperature of the electronic device **200** is converted into the electrical characteristic value.

[0097] According to an embodiment, the communication processor **220** may acquire the information related to the temperature of the electronic device **200** from the application processor **210**. In an example, the application processor **210** may transmit the information related to the temperature of the electronic device **200** in response to the request of the communication processor **220**.

[0098] According to an embodiment, the communication processor **220** may determine the second parameter of the transmission power as the first level, the second level, the third level, and/or the fourth level on the basis of the temperature of the electronic device **200**. The first level, the second level, the third level, and/or the fourth level of the second parameter may be the value related to the transmission power of the designated value for lowering the transmission power.

[0099] In operation **521**, the communication processor **220** may identify whether the temperature of the electronic device **200** is higher than or equal to a first value.

[0100] According to an embodiment, the first value may be about 41.5°.

[0101] According to various embodiments, in operation **531**, the communication processor **220** may control the RF module **230** by setting the second parameter of the transmission power as the first level in response to the temperature of the electronic device **200** being less than the first value (e.g., operation **521**—No).

[0102] The first level of the second parameter of the transmission power may be a value set as a default in the RF module **230** of the electronic device **200**.

[0103] According to various embodiments, in operation **522**, the communication processor **220** may identify whether the temperature of the electronic device **200** is higher than or equal to a second value in response to the temperature of the electronic device **200** being higher than or equal to the first value (e.g., operation **521**—Yes).

[0104] According to another embodiment, the second value may be about 43°.

[0105] In operation **532**, the communication processor **220** may control the RF module **230** by setting the second parameter of the transmission power as the second level in response to the temperature of the electronic device **200** being higher than or equal to the first value and less than the second value (e.g., operation **522**—No).

[0106] According to another embodiment, the second level of the second parameter of the transmission power may be transmission power having a lower value than the first level.

[0107] According to various embodiments, in operation **523**, the communication processor **220** may identify whether the temperature of the electronic device **200** is higher than or equal to a third value in response to the temperature of the electronic device **200** being higher than or equal to the second value (e.g., operation **522**—Yes).

[0108] In operation **533**, the communication processor **220** may control the RF module **230** by setting the second parameter of the transmission power as the third level in response to the temperature of the electronic device **200** being higher than or equal to the second value and less than the third value (e.g., operation **523**—No).

[0109] According to an embodiment, the third value may be about 45°.

[0110] According to an embodiment, the third level of the second parameter of the transmission power may be transmission power having a lower value than the second level.

[0111] According to various embodiments, in operation **534**, the communication processor **220** may control the RF module **230** by setting the second parameter of the transmission power as the fourth level in response to the temperature of the electronic device **200** being higher than or equal to the third value (e.g., operation **523**—Yes).

[0112] According to another embodiment, the fourth level of the second parameter of the transmission power may be transmission power having a lower value than the third level.

[0113] According to various embodiments, the communication processor **220** may perform operations **510** to **534** every designated cycle.

[0114] FIG. **5B** is a diagram illustrating a transmission power graph when an electronic device **200** controls an RF module **230** on a basis of a temperature of an electronic device **200** according to an embodiment of the disclosure.

[0115] In the TAS control, the parameter related to the transmission power may be applied. For example, the parameter related to the transmission power may include the first parameter, the second parameter, and/or the third parameter.

[0116] For example, the first parameter of the transmission power is the threshold value P_{limit} of the transmission power for satisfying the target SAR for the designated period of time, which may be the maximum time-average output power.

[0117] For example, the second parameter of the transmission power may be the value for outputting the designated power to lower the transmission power when the transmission power exceeds the designated value.

[0118] The third parameter of the transmission power may be the maximum power P_{max} that the transmitter of the RF module **230** may output.

[0119] For example, the target SAR may be the reference SAR for satisfying the TAS.

[0120] In another example, the averaged SAR may be the averaged SAR for the designated period of time.

[0121] The electronic device **200** may control the RF module **230** on the basis of the first

parameter, the second parameter, and/or the third parameter so that the averaged SAR of the transmission power for the designated period of time satisfies the target SAR.

[0122] According to various embodiments, the communication processor **220** may acquire the information related to the temperature of the electronic device **200** according to operation **510**.

[0123] According to other embodiments, the temperature of the electronic device **200** may be less than the first value before time point A.

[0124] According to various embodiments, the communication processor **220** may control the RF module **230** by setting the second parameter of the transmission power as the first level according to operation **531** in response to the temperature of the electronic device **200** being less than the first value (e.g., operation **521**—No).

[0125] According to some embodiments, at time point “A”, the temperature of the electronic device **200** may be higher than or equal to the first value and equal to or less than the second value.

[0126] According to various embodiments, the communication processor **220** may control the RF module **230** by setting the second parameter of the transmission power as the second level according to operation **532** in response to the temperature of the electronic device **200** being higher than or equal to the first value and less than the second value (e.g., operation **522**—No).

[0127] Accordingly, the communication module may maintain the averaged SAR value of the transmission power to be the target or less.

[0128] FIG. **6A** is a flowchart illustrating a method for an electronic device **200** to control an RF module **230** on a basis of a temperature of an electronic device **200** according to an embodiment of the disclosure.

[0129] According to some embodiments, in operation **610**, the communication processor **220** may acquire the information related to the temperature of the electronic device **200**.

[0130] According to an embodiment, the temperature sensor **260** may measure the temperature of the electronic device **200**. For example, the temperature sensor **260** is a component for converting the temperature into the electrical characteristic value, and may include at least one of the thermistor, the resistance thermometer, the thermoelectric, the silicon transducer, or the critical temperature resistor (CTR). The temperature sensor **260** may include at least one of various other temperature detection elements.

[0131] According to another embodiment, the application processor **210** may acquire the information related to the temperature of the electronic device **200** from the temperature sensor **260**. For example, the application processor **210** may request the information related to the temperature of the electronic device **200** from the temperature sensor **260** and acquire, from the temperature sensor **260**, the information in which the temperature of the electronic device **200** is converted into the electrical characteristic value.

[0132] The communication processor **220** may acquire the information related to the temperature of the electronic device **200** from the application processor **210**. For example, the application processor **210** may transmit the information related to the temperature of the electronic device **200** in response to the request of the communication processor **220**.

[0133] According to various embodiments, in operation **621**, the communication processor **220** may identify whether the temperature of the electronic device **200** is higher than or equal to a fourth value.

[0134] According to another embodiment, the fourth value may be about 38°.

[0135] According to various embodiments, in operation **631**, the communication processor **220** may control the RF module **230** by setting the first parameter of the transmission power as the first level in response to the temperature of the electronic device **200** being less than the fourth value (e.g., operation **621**—No).

[0136] The first level of the first parameter of the transmission power may be a value that sets the target SAR to 1 W/Kg as a reference.

[0137] According to an embodiment, in order to set the first parameter of the transmission power as

the first level, the communication processor **220** may set the time for maintaining the transmission power at a second parameter value as a first time (e.g., 50 ms).

[0138] In operation **622**, the communication processor **220** may identify whether the temperature of the electronic device **200** is higher than or equal to a fifth value in response to the temperature of the electronic device **200** being higher or equal to the fourth value (e.g., operation **621**—Yes).

[0139] According to an embodiment, the fifth value may be about 40°.

[0140] According to various embodiments, in operation **632**, the communication processor **220** may control the RF module **230** by setting the first parameter of the transmission power as the second level in response to the temperature of the electronic device **200** being higher than or equal to the fourth value and less than the fifth value (e.g., operation **622**—No).

[0141] According to another embodiment, the second level of the first parameter of the transmission power may be a value that sets the target SAR to 0.9 W/Kg as a reference.

[0142] According to an embodiment, in order to set the first parameter of the transmission power as the second level, the communication processor **220** may set the time for maintaining the transmission power at the second parameter value as a second time (e.g., 80 ms).

[0143] According to some embodiments, in operation **633**, the communication processor **220** may control the RF module **230** by setting the first parameter of the transmission power as the third level in response to the temperature of the electronic device **200** being higher than or equal to the fifth value (e.g., operation **622**—Yes).

[0144] According to another embodiment, the third level of the first parameter of the transmission power may be a value that sets the target SAR to 0.8 W/Kg as a reference.

[0145] In order to set the first parameter of the transmission power as the third level, the communication processor **220** may set the time for maintaining the transmission power at the second parameter value as a third time (e.g., 100 ms).

[0146] According to various embodiments, the communication processor **220** may repeatedly perform operations **610** to **633** every designated cycle.

[0147] FIG. **6B** is a diagram illustrating a transmission power graph when an electronic device **200** controls an RF module **230** on a basis of a temperature of an electronic device **200** according to an embodiment of the disclosure.

[0148] In the TAS control, the parameter related to the transmission power may be applied. For example, the parameter related to the transmission power may include the first parameter, the second parameter, and/or the third parameter.

[0149] For example, the first parameter of the transmission power is the threshold value P_{limit} of the transmission power for satisfying the target SAR for the designated period of time, which may be maximum time-average output power.

[0150] In another example, the second parameter of the transmission power may be the value for outputting the designated power to lower the average transmission power when the transmission power exceeds the designated value.

[0151] For example, the third parameter of the transmission power may be the maximum power P_{max} that the transmitter of the RF module **230** may output.

[0152] The target SAR may be the reference SAR for satisfying the TAS.

[0153] For example, the averaged SAR may be the averaged SAR for the designated period of time.

[0154] The electronic device **200** may control the RF module **230** on the basis of the first parameter, the second parameter, and/or the third parameter so that the averaged SAR of the transmission power for the designated period of time satisfies the target SAR.

[0155] According to various embodiments, the communication processor **220** may acquire the information related to the temperature of the electronic device **200** according to operation **610**.

[0156] The temperature of the electronic device **200** may be less than the fourth value before the time point A.

[0157] According to various embodiments, the communication processor **220** may control the RF module **230** by setting the first parameter of the transmission power as the first level according to operation **631** in response to the temperature of the electronic device **200** being less than the fourth value (e.g., operation **621**—No).

[0158] According to some embodiments, at time point A, the temperature of the electronic device **200** may be higher than or equal to the fourth value and less than the fifth value.

[0159] According to various embodiments, the communication processor **220** may control the RF module **230** by setting the first parameter of the transmission power as the second level according to operation **632** in response to the temperature of the electronic device **200** being higher or equal to the fourth value and less than the fifth value (e.g., operation **622**—No).

[0160] Accordingly, the communication module may lower the averaged SAR of the transmission power.

[0161] FIG. 7A is a flowchart illustrating a method for an electronic device **200** to control an RF module **230** on a basis of a temperature of an electronic device **200** according to an embodiment of the disclosure.

[0162] According to various embodiments, in operation **710**, the communication processor **220** may acquire the information related to the temperature of the electronic device **200**.

[0163] According to an embodiment, the temperature sensor **260** may measure the temperature of the electronic device **200**. For example, the temperature sensor **260** is a component for converting the temperature into the electrical characteristic value, and may include at least one of the thermistor, the resistance thermometer, the thermoelectric, the silicon transducer, or the critical temperature resistor (CTR). The temperature sensor **260** may include at least one of various other temperature detection elements.

[0164] According to another embodiment, the application processor **210** may acquire the information related to the temperature of the electronic device **200** from the temperature sensor **260**. For example, the application processor **210** may request the information related to the temperature of the electronic device **200** from the temperature sensor **260** and acquire, from the temperature sensor **260**, the information in which the temperature of the electronic device **200** is converted into the electrical characteristic value.

[0165] The communication processor **220** may acquire the information related to the temperature of the electronic device **200** from the application processor **210**. For example, the application processor **210** may transmit the information related to the temperature of the electronic device **200** in response to the request of the communication processor **220**.

[0166] According to various embodiments, in operation **720**, the communication processor **220** may identify whether the temperature of the electronic device **200** is higher than or equal to the designated value.

[0167] According to various embodiments, in operation **731**, the communication processor **220** may set the maximum value of the transmission power as a third parameter value in response to the temperature of the electronic device **200** being less than the designated value (e.g., operation **720**—No).

[0168] In operation **732**, the communication processor **220** may set the maximum value of the transmission power as a first parameter value in response to the temperature of the electronic device **200** being higher than or equal to the designated value (e.g., operation **720**—Yes).

[0169] According to various embodiments, the communication processor **220** may repeatedly perform operations **710** to **732** every designated cycle.

[0170] FIG. 7B is a diagram illustrating a transmission power graph when an electronic device **200** controls an RF module **230** on a basis of a temperature of an electronic device **200** according to an embodiment of the disclosure.

[0171] In the TAS control, the parameter related to the transmission power may be applied. For example, the parameter related to the transmission power may include the first parameter, the

second parameter, and/or the third parameter.

[0172] For example, the first parameter of the transmission power is the threshold value P_{limit} of the transmission power for satisfying the target SAR for the designated period of time, which may be maximum time-average output power.

[0173] For example, the second parameter of the transmission power may be the value for outputting the designated power to lower the average transmission power when the transmission power exceeds the designated value.

[0174] The third parameter of the transmission power may be the maximum power P_{max} that the transmitter of the RF module **230** may output.

[0175] For example, the target SAR may be the reference SAR for satisfying the TAS.

[0176] For example, the averaged SAR may be the averaged SAR for the designated period of time.

[0177] The electronic device **200** may control the RF module **230** on the basis of the first parameter, the second parameter, and/or the third parameter so that the averaged SAR of the transmission power for the designated period of time satisfies the target SAR.

[0178] According to various embodiments, the communication processor **220** may acquire the information related to the temperature of the electronic device **200** according to operation **710**.

[0179] According to other embodiments, the temperature of the electronic device **200** may be less than the designated value before the time point A.

[0180] According to various embodiments, the communication processor **220** may control the RF module **230** by setting the maximum value of the transmission power as the third parameter value according to operation **731** in response to the temperature of the electronic device **200** being less than the designated value (e.g., operation **720**—No).

[0181] The temperature of the electronic device **200** may be higher than or equal to the designated value at the time point A.

[0182] According to various embodiments, the communication processor **220** may control the RF module **230** by setting the maximum value of the transmission power as the first parameter value according to operation **732** in response to the temperature of the electronic device **200** being higher than or equal to the designated value (e.g., operation **720**—No).

[0183] Accordingly, the communication module may maintain the averaged SAR value of the transmission power to be the target or less.

[0184] An electronic device according to various embodiments provided in the disclosure includes a temperature sensor, an application processor, a communication processor and an RF module, in which the communication processor may acquire, from the application processor, information related to the electronic device temperature measured by the temperature sensor, determine a parameter related to the transmission power of the RF module on the basis of the temperature of the electronic device, and control the RF module on the basis of the determined parameter.

[0185] According to an embodiment, the parameter related to the transmission power may include a first parameter that is a threshold value of the transmission power for satisfying the target SAR for the designated period of time, and the communication processor may determine the first parameter on the basis of the temperature of the electronic device.

[0186] According to another embodiment, the communication processor may determine the first parameter as the first level in response to the temperature of the electronic device being less than the designated value, and determine the first parameter as the second level in response to the temperature of the electronic device being higher than or equal to the designated value. The second level may mean a threshold value lower than the first level.

[0187] The communication processor may determine, at the first level of the first parameter, the time when the RF module outputs the designated power to lower the average transmission power as the first time, and at the second level of the first parameter, the time when the RF module outputs the designated power to lower the average transmission power as the second time. The second time

may be longer than the first time.

[0188] According to an embodiment, the parameter related to the transmission power includes the second parameter that is a value for outputting the designated power to lower the average transmission power when the transmission power exceeds the designated value, and the communication processor may determine the second parameter on the basis of the temperature of the electronic device.

[0189] The communication processor determines the second parameter as the first level in response to the temperature of the electronic device being less than the designated value, and determines the second parameter as the second level in response to the temperature of the electronic device being higher than or equal to the designated value, and the second level may mean the transmission power having a value lower than the first level.

[0190] According to an embodiment, the parameter related to the transmission power includes the third parameter that is the maximum power value that the transmitter of the RF module may output, and the communication processor may determine the third parameter on the basis of the temperature of the electronic device.

[0191] According to another embodiment, the communication processor determines the third parameter as the first level in response to the temperature of the electronic device being less than the designated value, and determines the third parameter as the second level in response to the temperature of the electronic device being higher than or equal to the designated value, and the second level may mean the transmission power having a value lower than the first level.

[0192] The electronic device operating method according to an embodiment may include acquiring the information related to the temperature of the electronic device measured by the temperature sensor from the application processor, determining the parameter related to the transmission power of the RF module on the basis of the temperature of the electronic device, and controlling the RF module on the basis of the determined parameter.

[0193] It will be appreciated that various embodiments of the disclosure according to the claims and description in the specification can be realized in the form of hardware, software or a combination of hardware and software.

[0194] Any such software may be stored in non-transitory computer readable storage media. The non-transitory computer readable storage media store one or more computer programs (software modules), the one or more computer programs include computer-executable instructions that, when executed by one or more processors of an electronic device individually or collectively, cause the electronic device to perform a method of the disclosure.

[0195] Any such software may be stored in the form of volatile or non-volatile storage such as, for example, a storage device like read only memory (ROM), whether erasable or rewritable or not, or in the form of memory such as, for example, random access memory (RAM), memory chips, device or integrated circuits or on an optically or magnetically readable medium such as, for example, a compact disk (CD), digital versatile disc (DVD), magnetic disk or magnetic tape or the like. It will be appreciated that the storage devices and storage media are various embodiments of non-transitory machine-readable storage that are suitable for storing a computer program or computer programs comprising instructions that, when executed, implement various embodiments of the disclosure. Accordingly, various embodiments provide a program comprising code for implementing apparatus or a method as claimed in any one of the claims of this specification and a non-transitory machine-readable storage storing such a program.

[0196] While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. An electronic device, comprising: a temperature sensor; an application processor; a radio frequency (RF) module, memory, comprising one or more storage media, storing instructions; and one or more communication processors communicatively coupled to the temperature sensor, the application processor, the RF module, and the memory, wherein the instructions, when executed by the one or more communication processors individually or collectively, cause the electronic device to: acquire, from the application processor, information related to a temperature of the electronic device measured by the temperature sensor, determine a parameter related to transmission power of the RF module based on the temperature of the electronic device, and control the RF module based on the determined parameter.
2. The electronic device of claim 1, wherein the parameter related to the transmission power include a first parameter which is a threshold value of the transmission power for satisfying a target specific absorption rate (SAR) for a designated period of time, and wherein the instructions, when executed by the one or more communication processors individually or collectively, further cause the electronic device to determine the first parameter based on the temperature of the electronic device.
3. The electronic device of claim 2, wherein the instructions, when executed by the one or more communication processors individually or collectively, further cause the electronic device to: determine the first parameter as a first level in response to the temperature of the electronic device being less than a designated value, and determine the first parameter as a second level in response to the temperature of the electronic device being higher than or equal to the designated value, and wherein the second level is a threshold value lower than the first level.
4. The electronic device of claim 3, wherein the instructions, when executed by the one or more communication processors individually or collectively, further cause the electronic device to: determine, at the first level of the first parameter, a time when which the RF module outputs designated power to lower average transmission power as a first time, and determine, at the second level of the first parameter, a time when which the RF module outputs the designated power to lower the average transmission power as a second time, and wherein the second time is longer than the first time.
5. The electronic device of claim 1, wherein the parameter related to the transmission power includes a second parameter that is a value for outputting designated power to lower average transmission power when the transmission power exceeds a designated value, and wherein the instructions, when executed by the one or more communication processors individually or collectively, further cause the electronic device to determine the second parameter based on the temperature of the electronic device.
6. The electronic device of claim 5, wherein the instructions, when executed by the one or more communication processors individually or collectively, further cause the electronic device to: determine the second parameter as a first level in response to the temperature of the electronic device being less than the designated value, and determine the second parameter as a second level in response to the temperature of the electronic device being higher than or equal to the designated value, and wherein the second level is transmission power having a value lower than the first level.
7. The electronic device of claim 1, wherein the parameter related to the transmission power includes a third parameter that is a maximum power value that a transmitter of the RF module outputs, and wherein the instructions, when executed by the one or more communication processors individually or collectively, further cause the electronic device to determine the third parameter based on the temperature of the electronic device.
8. The electronic device of claim 7, wherein the instructions, when executed by the one or more communication processors individually or collectively, further cause the electronic device to:

determine the third parameter as a first level in response to the temperature of the electronic device being less than a designated value, and determine the third parameter as a second level in response to the temperature of the electronic device being higher than or equal to the designated value, and wherein the second level is transmission power having a value lower than the first level.

9. A method of operating an electronic device, the method comprising: acquiring information related to a temperature of an electronic device measured by a temperature sensor from an application processor; determining a parameter related to transmission power of an radio frequency (RF) module based on the temperature of the electronic device; and controlling the RF module based on the determined parameter.

10. The method of claim 9, wherein the parameter related to the transmission power includes a first parameter which is a threshold value of the transmission power for satisfying a target specific absorption rate (SAR) for a designated period of time, and wherein the method further comprises: determining the first parameter based on the temperature of the electronic device.

11. The method of claim 10, further comprising: determining the first parameter as a first level in response to the temperature of the electronic device being less than a designated value; and determining the first parameter as a second level in response to the temperature of the electronic device being higher than or equal to the designated value, wherein the second level is a threshold value lower than the first level.

12. The method of claim 11, further comprising: determining, at the first level of the first parameter, a time when which the RF module outputs designated power to lower average transmission power as a first time; and determining, at the second level of the first parameter, a time when which the RF module outputs designated power to lower the average transmission power as a second time, wherein the second time is longer than the first time.

13. The method of claim 9, wherein the parameter related to the transmission power includes a second parameter that is a value for outputting designated power to lower average transmission power when the transmission power exceeds the designated value, and wherein the method further comprises: determining the second parameter based on the temperature of the electronic device.

14. The method of claim 13, further comprising: determining the second parameter as a first level in response to the temperature of the electronic device being less than the designated value; and determining the second parameter as a second level in response to the temperature of the electronic device being higher than or equal to the designated value, wherein the second level is transmission power having a value lower than the first level.

15. The method of claim 9, wherein the parameter related to the transmission power includes a third parameter that is a maximum power value that a transmitter of the RF module outputs, and wherein the method further comprises: determining the third parameter based on the temperature of the electronic device.

16. The method of claim 15, further comprising: determining the third parameter as a first level in response to the temperature of the electronic device being less than a designated value; and determining the third parameter as a second level in response to the temperature of the electronic device being higher than or equal to the designated value, wherein the second level is transmission power having a value lower than the first level.

17. One or more non-transitory computer-readable storage media storing one or more computer programs including computer-executable instructions that, when executed by one or more communication processors of an electronic device individually or collectively, cause the electronic device to perform operations, the operations comprising: acquiring information related to a temperature of an electronic device measured by a temperature sensor from an application processor; determining a parameter related to transmission power of an radio frequency (RF) module based on the temperature of the electronic device; and controlling the RF module based on the determined parameter.

18. The one or more non-transitory computer-readable storage media of claim 17, wherein the

parameter related to the transmission power includes a first parameter which is a threshold value of the transmission power for satisfying a target specific absorption rate (SAR) for a designated period of time, and wherein the operations further comprise: determining the first parameter based on the temperature of the electronic device.
